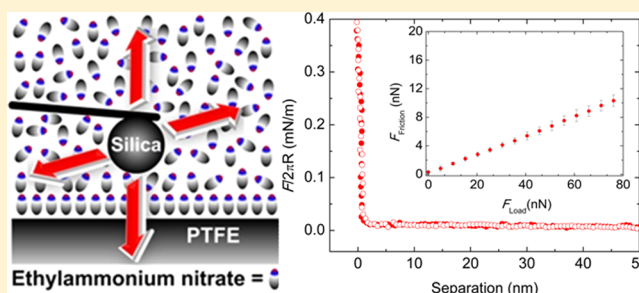


Ionic Liquid Nanotribology: Stiction Suppression and Surface Induced Shear Thinning

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Supporting Information

ABSTRACT: The friction and adhesion between pairs of materials (silica, alumina, and polytetrafluoroethylene) have been studied and interpreted in terms of the long-ranged interactions present. In ambient laboratory air, the interactions are dominated by van der Waals attraction and strong adhesion leading to significant frictional forces. In the presence of the ionic liquid (IL) ethylammonium nitrate (EAN) the van der Waals interaction is suppressed and the attractive/adhesive interactions which lead to “stiction” are removed, resulting in an at least a 10-fold reduction in the friction force at large applied loads. The friction coefficient for each system was determined; coefficients obtained in air were significantly larger than those obtained in the presence of EAN (which ranged between 0.1 and 0.25), and variation in the friction coefficients between systems was correlated with changes in surface roughness. As the viscosity of ILs can be relatively high, which has implications for the lubricating properties, the hydrodynamic forces between the surfaces have therefore also been studied. The linear increase in repulsive force with speed, expected from hydrodynamic interactions, is clearly observed, and these forces further inhibit the potential for stiction. Remarkably, the viscosity extracted from the data is dramatically reduced compared to the bulk value, indicative of a surface ordering effect which significantly reduces viscous losses.



INTRODUCTION

The emergence of micro- and nanoelectromechanical systems (MEMS/NEMS) over the past two decades has been made possible by new technologies for fabricating, assembling, and controlling smaller and smaller components.¹ Unfortunately, the microscopic length scale and high surface-area-to-volume ratio make such devices prone to adhesion, friction, and wear, specifically when movable parts come into contact.^{2,3} These tribological phenomena make it extremely difficult to meet the precise operating standards required over practical lifetimes and are the major reasons preventing a broader use of sliding contacts in MEMS and NEMS.^{3,4}

The role of a lubricant is to reduce friction and dissipate heat. Conventional lubricants, used at high shear rates achieve this via a hydrodynamic mechanism which prevents the surfaces from contacting. For boundary lubrication (where surface contact is unavoidable, for example at low shear rates) this can be done either by reducing the adhesion and attractive forces that contribute to friction (this is usually done with boundary lubricant additives which adsorb and form a palisade layer) or by reducing the dependence of friction on the normal load (i.e., reducing the friction coefficient). The choice of lubricant will depend on parameters such as the shear rate, the chemistry of the materials being lubricated, whether the system is open or

closed to the environment, the working temperature, electrical currents, and shelf life of the device, to name just a few. A lubricant must also be easy to apply, environmentally stable, highly durable and ideally form a molecularly thick layer which is strongly bound to the surface.^{1,5} Ionic liquids (ILs) meet many of these requirements and have received considerable attention in recent times as potential lubricants, especially for MEMS and NEMS as they are inherently conductive. Their strong ordering at the interfaces,^{6–12} as well as the possibility of tuning the viscous behavior, renders them interesting as “multifunctional lubricants” they could potentially act as both boundary and elastohydrodynamic lubricants.

ILs are molten organic salts with melting temperatures below 100 °C. ILs often have high temperature stability and thermal conductivity, negligible volatility, and electrical conductivity which is useful for lubrication applications where current flow is required between sliding contacts.² The physical properties of ILs can be tuned easily by selecting appropriate combinations of ions¹⁰ which has led to applications including green solvents/catalysts,^{13–15} electrolytes in solar cells and batteries,¹⁶

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as solvents for self-assembly^{17,18} and polymers,¹⁹ separation and purification technologies,²⁰ particle stabilization,¹⁰ heat transfer fluids,²¹ specialty applications such as reprocessing of nuclear waste,²² and, more recently, as lubricants.^{1,23–30}

One of the most widely studied ILs is ethylammonium nitrate (EAN), comprised of an ethyl ammonium cation and a nitrate anion.^{5,10} EAN, discovered in 1914,¹⁰ is a protic IL with some water-like properties resulting from the ability of EAN to form a 3D hydrogen bonded network, but its viscosity is 30 times higher than water.¹² EAN self-organizes into a lamellar-like structure near a flat surface,^{8,9,31} decaying into a disordered sponge phase away from the interface.^{7,32} This structuring may enhance the inherent lubrication properties of ILs. EAN can be considered a “green lubricant” on many levels: (1) its synthesis is green (high yield, water is the only solvent, minimal waste, and low energy consumption), (2) when used as a lubricant its nonvolatility leads to reduced air emissions when compared to hydrocarbon-based oil lubricants, (3) its durability/stability suggests it could last the lifetime of the device without needing to be replaced or exchanged, (4) it is relatively nontoxic,³³ and (5) it would allow devices to operate more efficiently and last longer with conserved complexity. As such, IL lubricants fulfill the requirement of being more environmentally benign than existing technologies and less expensive than comparable conventional lubricants such as perfluoropolyethers.¹

While the atomic force microscope (AFM) is often used to image nanometer-scale topography, it has the piconewton force resolution required to measure the forces associated with liquid molecules near surfaces by recording the deflection of a microscale cantilever.³⁴ Attaching a colloidal particle (1–20 μm in diameter) to a tipless cantilever (instead of using a sharp tip with radius ca. 10 nm) has the advantage of being able to measure forces with a probe of virtually any material as long as its shape is well-defined and relatively incompressible.³⁵ This renders the colloidal probe AFM technique ideal for studies of nanotribology.^{36,37}

Following pioneering surface forces apparatus (SFA) studies of Horn et al. concerning EAN more than 20 years ago,³⁸ Atkin and co-workers have investigated IL structural/solvation forces in ILs via AFM normal force measurements using AFM tips^{6–11} and colloid probes;^{10,12} similar results were obtained for both probe types. Oscillatory forces were measured in a variety of ILs where the oscillation period corresponded to the physical dimensions of the cation–anion pair and the height of the oscillation indicated the force necessary to “push through” ion layers.

Lateral AFM measurements have also been performed in a range of ILs using conventional cantilevers with Si₃N₄ tips to measure friction. For example, ethylammonium nitrate,^{8,9,11} propylammonium nitrate,⁹ imidazolium,^{7,9,28,39} and pyrrolidinium^{6,7} type ILs. In most cases, the nominal dimensions of these tips and cantilevers, as well as the spring constants, provided by the manufacturer were used in data processing. Such values however, are often subject to large deviations due to defects and batch-to-batch variations which can affect the values returned and render comparison difficult.^{40,41} For this reason, it is desirable to measure cantilever/probe dimensions and calibrate spring constants experimentally, for example using the method of Sader.⁴²

Recently, colloid probe AFM friction force measurements have examined a variety of aprotic ILs, including ethyl methyl pyridinium ethyl sulfate, 1-propyl-3-methylimidazolium bromide and 1-ethanol-3-methylimidazolium chloride.^{4,43,44} Addi-

tionally, previous normal force measurements have studied structural changes in ILs at the solid–liquid interface.^{10,12} The surface force apparatus (SFA)^{38,43,45–47} has also been used for force measurements, enabling structural order, viscosity and shear properties to be determined; the drawback of this technique, however, is that layering confinement can only be measured in specific systems such as silica–silica⁴³ or mica–mica^{38,45,46} for reasons of optical transparency.

In this study, colloid probe AFM is used to measure the normal and frictional forces for several surfaces in air and EAN. The experiments are performed with sliding speeds and applied loads that are representative of MEMS/NEMS systems. van der Waals, electrostatic, and capillary forces are suppressed in EAN for a range of industrially relevant materials (with relatively high surface roughnesses); specifically, silica and alumina probes against silica and polytetrafluoroethylene (PTFE) surfaces have been investigated. We demonstrate that although intersurface forces are strongly material dependent in air, in EAN, normal forces become negligible and essentially material independent, which leads to a significant reduction in stiction (a now commonly used composite of “sticking” and “friction”).

MATERIALS AND METHODS

Materials. Ethylamine (68% w/w) was purchased from Sigma Aldrich (Munich, Germany), and nitric acid (65% w/w) was acquired from Merck (Darmstadt, Germany). Deionized water (18.2 M Ω cm resistivity) was obtained from a Millipore Milli-Q Purification System (Millipore, Malsheim, France), and pure ethanol (99.5%) was acquired from Kemetyl (Haninge, Sweden). The two component epoxy glue for gluing colloid probes (Araldite Rapid, Newtown, Ireland) was acquired from Huntsman LLC (Duxford, United Kingdom), and chromosulfuric acid for cleaning substrates (BIC) was purchased from Merck (Darmstadt, Germany).

Cantilever Calibration. Cantilevers were chosen based on the expected magnitude of adhesive/attractive forces, where a lower spring constant can be used to detect a weaker force. Tipless cantilevers (CSC12 no backside coating) were purchased from MikroMasch (Tallinn, Estonia) with normal spring constants ranging from 0.15 to 1.5 N/m. The cantilevers were calibrated using the method developed by Sader.⁴⁸ Before the attachment of the particle to the cantilever, the normal Q value and normal resonant frequency were determined by the thermal noise method using the Tune-IT v. 2.6.1 software (ForceIT, Sweden). The cantilever dimensions were measured by optical microscopy (Nikon Optiphot 100, Tokyo, Japan) and digitally analyzed using the open-source software ImageJ (NIST, Gaithersburg, MD). These values were combined to calculate the normal spring constants (k_z) using the Tune-IT software. The particles, either silica, SiO₂, (Bangs Laboratories, Fishers, IN) or alumina, Al₂O₃, (generously provided by YKI, Stockholm) were then glued onto the CSC12 cantilevers. An example of one of the colloid probes used in this study is shown in Figure S3.

Conversely, the torsional spring constants (k_ϕ), needed for friction measurement data processing, were calculated from the normal spring constant (k_z), dimensions, and material parameters according to eq 1⁴⁹

$$k_\phi = \frac{k_z 4L^3}{6(1+\nu)(L-\Delta L)} \left\{ 1 - \left(\frac{\tanh\left(\frac{L-\Delta L}{w} \sqrt{6(1-\nu)}\right)}{\sqrt{6(1-\nu)}} \frac{w}{(L-\Delta L)} \right) \right\}^{-1} \quad (1)$$

where ν is Poisson's ratio, L and w are the length and width of the cantilever, and ΔL is the distance from the center of the glued particle to the cantilever's free end.

AFM Imaging: Surface Roughness Measurements. Morphology of the particles glued on the cantilevers and the planar surfaces

used for force measurements were imaged using a NanoScope IIIa Atomic Force Microscope (Veeco, Santa Barbara, CA) with the J piezoelectric scanner. Images were collected in tapping mode using Ultrasharp Si tips (NSC14/ALBS, MikroMasch, Tallinn, Estonia, with normal spring constants of 5.5 N/m). Before imaging or force measurements, glued probes were cleaned by thorough rinsing with water and ethanol, followed by drying with N₂ gas and then plasma treatment (Sterilizer PDC-32G, Harrick, Ossining, NY) for 3 min. Silica wafers (Ultrapack Wafershield, H9100-0302, Entegris, Dresden, Germany) were cut to the desired dimensions (ca. 1 cm × 1 cm) and cleaned in a BIC (bichromosulfuric acid) bath for 0.5 h, followed by extensive rinsing with water and ethanol. The cleaned wafers were dried with N₂ and plasma cleaned for 3 min as well. Smooth PTFE surfaces were produced by compressing pieces of PTFE between freshly cleaved atomically flat mica sheets in an oven at 500 °C overnight,⁵⁰ then cleaning with water and ethanol, and drying with N₂ before measurements. Root-mean-square (rms) roughness was calculated by averaging 12 values from 1 μm² areas on the AFM 2 × 2 μm images using the AFM software (Nanoscope 6.13R1).

AFM Force and Friction Measurements. Force and friction measurements were performed in air and in ethylammonium nitrate (EAN) that was synthesized in-house, as described previously,¹⁴ from ethylamine and nitric acid. Interactions were investigated in the following systems (using the convention probe-surface): SiO₂-SiO₂, SiO₂-PTFE, Al₂O₃-SiO₂, and Al₂O₃-PTFE using a Multimode Picoforce AFM (Veeco, Santa Barbara, CA) equipped with a closed looped Picoforce scanner. The standard liquid cell was employed but solutions were nonetheless exposed to laboratory atmospheric conditions. All of the data from AFM measurements were processed using the ForceIT v. 2.6.1 software (ForceIT, Sweden). For comparison purposes, measured forces between the spherical probe and planar surfaces were normalized by 2πR, where R is the radius of the particle. According to the Derjaguin approximation,³⁷ the normalized forces are equal to the interaction energy per unit area between two planes. For normal force measurements, force pulls with ramp sizes of either 500 nm or 2 μm at different rates (0.1–3 Hz) were triggered at 200 nm deflection. A total of 25–30 curves were collected in different locations on each surface. The number of data points in the force curves shown in Figures 1, 2, 5, and S2 were reduced by a factor of 10 for the clarity of the figure.

For the friction experiments, sliding friction was measured over a scan distance (line) of 10 or 5 μm at scan rates of 0.1–3 Hz (512 points/line, 16 lines). The applied load was varied in a loop, from 0 to 3 V and then unloaded from +3 to −3 V. Friction data was recorded at intervals of 0.2 V.

Error Calculations. In general, the error values (Δy) are confidence intervals calculated using eq 2 from the standard deviation

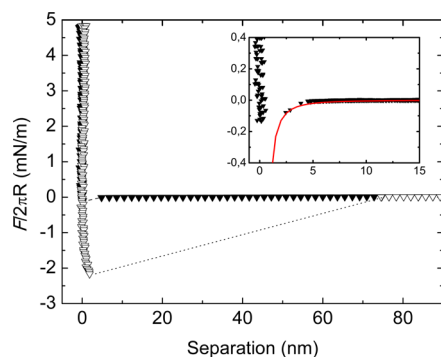


Figure 1. Interaction between silica-PTFE in air, normalized force curve on approach (closed symbols) and retraction (open symbols). The inset shows five normalized force curves on approach for the same system and the data fitted with van der Waals theory. The resulting Hamaker constant was 1.96×10^{-20} J (solid line). The jump-in (dashed line) and jump-out (dotted line) highlight the attraction and adhesion present in air, respectively.

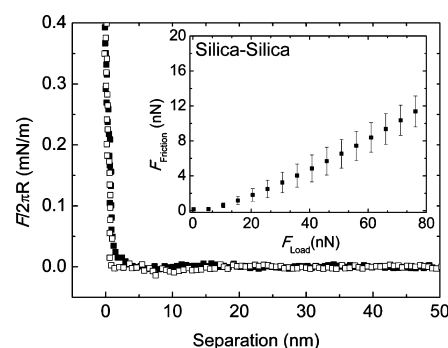


Figure 2. Normalized force curve on approach (closed symbols) and retraction (open symbols) at an approach rate of 100 nm/s and inset showing friction as a function of applied normal load in EAN for silica-silica. Almost identical results were obtained in the experiments using different materials (see Figure S2).

(σ_y) for a number of repeated measurements (N). In this case, t is Students' t -distribution value at a confidence level of 99.5% for $N - 1$ degree of freedom.

$$\Delta y = t\sigma_y/(N)^{1/2} \quad (2)$$

In friction experiments, the error was calculated from friction values at the same load during loading and unloading in repeated loops (they were recorded without cantilever withdrawal, one after the other). When N was lower than 3, the error bars shown are the standard deviation values. The standard deviation of the slope curve (to determine the error on μ) is used to calculate the error interval according to eq 2 using $N - 2$ degrees of freedom, which is conventional for linear regression error analysis.

Friction Coefficient. In 1699, Amontons postulated⁵¹ that friction between two materials is directly proportional to a normal applied load (F_L) but is independent of the apparent area of contact (in contrast to all other surface forces) according to eq 3

$$F_f(F_L) = \mu F_L \quad (3)$$

The proportionality between the lateral frictional force and the normal applied load is given by μ , termed the friction coefficient which is a useful parameter in comparing the lubrication properties between systems. When the adhesive forces in the system are on a comparable level to the applied load, the adhesion acts as an additional load and thus measurable friction extends to negative applied loads, and the "friction-load" relationship can often be simplified by the following equation (where $F_f(0)$ is the friction at zero applied load).³⁶

$$F_f(F_L) = \mu F_L + F_f(0) \quad (4)$$

By accounting for adhesion in this manner, a friction coefficient can thus be defined which is independent of adhesion (in the limit of nondeformable materials). Equation 4 is generally valid for both multiasperity contacts and adhesive contacts whenever deformation is small over the measured range.^{52,53} This empirical relationship has been proven in many experimental systems,^{54–57} has a broad range of applicability, and is the most common means of quantitatively describing the friction between surfaces.^{52,53,58}

RESULTS AND DISCUSSIONS

Force and friction measurements in four systems were studied by colloid probe AFM: silica-silica, silica-PTFE, alumina-silica, and alumina-PTFE. (The convention used in this work is probe-surface, so measurements denoted "silica-PTFE" are between a silica probe and a PTFE surface.) These materials were chosen according to two criteria. First, they are materials with practical significance and representative surface roughnesses (Table 1), for example, for MEMS/NEMS related devices. Second, it is possible to estimate the expected Hamaker

Table 1. Probe Dimensions and RMS Roughness for Surfaces and Probes Taken over $1 \times 1 \mu\text{m}$ Area AFM Images

	silica probe	alumina probe	silica surface	PTFE surface
rms roughness (nm)	17 ± 3	25 ± 3	0.60 ± 0.04	7 ± 1
Probe radius (μm)	4.4 ± 0.4	3.3 ± 0.2	N/A	N/A

constant from simplified Lifshitz theory based on the optical properties of the media.⁵⁹ In the presence of a medium with the same refractive index as EAN, the expected Hamaker constants for these combinations of materials would extend from negative to positive values, i.e., from attractive to repulsive van der Waals forces.^{50,60} However, a landmark paper³⁸ indicates that in the presence of an IL extensive screening of the interactions is to be expected. The four systems were analyzed in both ambient air and in EAN to understand how EAN modifies the forces and friction between the surfaces and thus how ILs might lubricate a contact.

Normalized force versus separation data for the silica–PTFE system in air are shown in Figure 1, and the corresponding friction dependence on the normal applied load is plotted in Figure S1. These figures provide an indication of the interactions in the absence of a solvent. On approach, a van der Waals attraction is observed, and on retraction, adhesive forces hold the surfaces together until they are pulled apart at large separations when the energy stored in the cantilever overcomes the adhesive interaction. The discontinuities in the force curve are highlighted with dotted and dashed lines: the jump-in observed on approach at short distances (below 5 nm) and the jump-out (around 77 nm) are characteristic of normal forces in air between a hydrophilic and a hydrophobic surface. Fitting of the approach curve using van der Waals theory gives a Hamaker constant of around 1.96×10^{-20} J (Figure 1, inset). This value is lower than the theoretical Hamaker constant for silica–PTFE in a nitrogen atmosphere (7.6×10^{-20} J⁶¹) which is not unexpected and results from a combination of the relatively high surface roughness in the silica–PTFE system (Table 1) and possibly also from the presence of an adsorbed water layer at the hydrophilic silica interface. The presence of water at the interface would shift the Hamaker constant toward that of silica–PTFE in water (0.36×10^{-20} J⁶²) and increased surface roughness has been shown to lead to a reduction of the contact area and consequently a reduced van der Waals interaction and Hamaker constant.⁶³ A further complication is the deformation of soft materials which is both positive (toward the approaching probe with an attractive interaction) and negative under adhesive load.⁶⁴

When silica–PTFE interactions in air are compared to measurements with the other systems in air (silica–silica, alumina–PTFE, and alumina–silica, data not shown) the **hydrophobicity** of PTFE makes this system representative of the minimal adhesion because of the absence of capillary forces. In the completely hydrophilic systems, strong capillary forces (and, at very low relative humidity, additional Coulombic forces due to tribocharging) lead to attractive forces and adhesion values which vary according to ambient conditions. (It follows that the friction values are also affected by the ambient conditions and for silica–silica the friction force is up to 3-times larger than those for silica–PTFE.) It has earlier been shown that the nanotribology of, for example, silica and mica is very dependent on the relative humidity, true area of contact, approach rate and time in contact.^{54,57,65}

The significant adhesion seen in Figure 1 leads to the large friction values observed in Figure S1. The friction coefficient is extracted from the gradient of the friction-load data in the linear regime and is thus not dependent on the magnitude of the load or frictional force; that is, the adhesion itself does not determine the friction coefficient. The extremely large friction at zero load is due to the significant adhesion acting as an additional load. Careful examination of Figures 1 and S1 reveals that the relationship between friction and load is not linear over the entire load regime. This is expected for a deformable contact and reflects the fact that the **true area of contact is increasing as the load increases**; such behavior is frequently observed for “single asperity-like” contacts.^{56,66} Thus the interactions in air are determined not only by the Hamaker constant of the system but also the surface energy and the relative humidity. (This would also be the case in a conventional liquid; in this case of course the Hamaker constant is generally reduced due to the dielectric properties of the liquid, and capillary bridges of water form only between hydrophilic surfaces in a nonpolar solvent.⁶⁷)

The immersion of the surfaces in a liquid medium is expected to completely change the nature of the interactions. In aqueous medium for example, there is a wealth of work describing the types of interactions that can be observed and the range of these surface forces is often greater than that of the van der Waals interaction in air. For the specific materials used here, experiments have shown ionic-strength dependent electrostatic double-layer forces on approach and adhesion due to van der Waals forces.^{62,68,69} Figure 2 shows that when the surface is immersed in IL all such surface forces are completely screened. The figure shows force versus separation data for the system silica–silica in EAN with its respective frictional force data shown as an inset. The systems silica–PTFE, alumina–silica, and alumina–PTFE in EAN are presented as Supporting Information in Figure S2 but are almost indistinguishable from the curve in Figure 2. No discernible van der Waals forces (or indeed any other type of surface force) were measured in EAN and little or no adhesion can be observed in either Figure 2 or Figure S2. Thus conventional surface forces no longer determine the surface interaction. With the alumina probe there is a small but discernible adhesion which is negligible when compared to the adhesion in air. This small attraction most likely results from the layering of EAN at short separations^{8,10,11,31,38} and potentially has two explanations. The first is that it is known that EAN orients with the alkane chains pointing away from an oppositely charged surface, and there is a solvophobic interaction.⁸ Second, since the interaction is not symmetric and the surfaces are furthermore not perfectly smooth, there is likely to be a mismatch between the ordered EAN layers on the probe and the surface which leads to localized Coulombic interactions between the two films. It is not possible to distinguish between these effects and both are expected to occur.

Friction measurements were made **over distances of $10 \mu\text{m}$ at a rate of $20 \mu\text{m/s}$** , approaching working conditions of MEMS such as micro and nanoactuators, which operate with sliding speeds of $5\text{--}50 \mu\text{m/s}$.^{70,71} (Note though that magnetic storage devices, for example, operate at vastly higher rates). Consistent with the absence of, or negligible, adhesion, every system studied in EAN displayed low friction with friction forces much smaller than for the same material combinations in air. Although the friction-load curves in air (Figure 1) followed a modified Amontons Law (eq 4), all friction curves in EAN

show zero friction at zero load and thus obey the more classical Amontons Law (eq 3). Moreover, for all of the samples studied, the frictional forces observed over the whole range of applied loads (0–80 nN) were below 30 nN (which is less than the friction in air at zero applied load!). The alumina–PTFE system presented the highest friction, probably because the alumina probe had the highest measured surface roughness among the four materials (and PTFE was the roughest of the planar surfaces). The friction coefficients determined from these measurements, are smaller in EAN than in air (see Figure 3) and are consistent with what is expected for conventional lubricants.³⁹

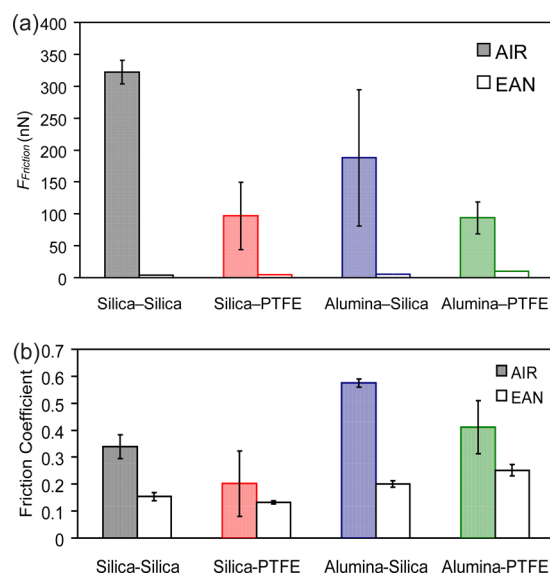


Figure 3. (a) Frictional force magnitude at an applied normal load of 35 nN for the four probe-surface combinations studied in air (hashed bars) and in EAN (open bars) and (b) friction coefficients.

Friction forces at an applied normal load of 35 nN (an arbitrary value chosen in the middle of the applied load range for comparison purposes) and friction coefficients are displayed in Figure 3 in order to provide an easier comparison of the results for the different material combinations. In air, frictional forces are strongly dependent on attraction and adhesion, which are highly influenced by the material's chemistry and hydrophobicity,⁵⁷ the environment (humidity and temperature), and the number/sequence of measurements. For polar surfaces in air (silica–silica and alumina–silica), adhesion from capillary forces⁵⁷ led to the highest friction values. The most evident effect of the IL is to greatly reduce the friction magnitude while moderately reducing the friction coefficient; in EAN, attraction and adhesion were negligible and the minimal friction observed was mainly a result of sample/probe roughness.

The adhesive forces measured in air varied considerably which manifests as lateral shifts in the friction versus applied load curves and leads to large error bars for the friction values,⁵⁴ as shown in the air data in Figure 3a. (Error bars in Figure 3a for the measurements in EAN are smaller than the line thickness and are thus imperceptible.) The large values of the friction in Figure 3a reflect the fact that the adhesion is very large and acts as an extra load. To an extent, these values may thus depend on the probe geometry and roughness. It has been shown that adhesion acts as an additional, internal load,^{53–58}

and this causes for example the nonzero friction at zero applied load observed in Figure S1. The absence of any adhesion in EAN thus causes the dramatic reduction in friction for Figure 3b. (Note that most liquids DO reduce the vdW force and thus some reduction would be expected in any liquid; however, in this case the absence of measurable van der Waals forces means that the adhesive contribution to the friction disappears completely.) Although the magnitude of friction increases as the adhesion increases (i.e., a larger y intercept in eq 4), the friction coefficient remains constant. The differences in friction coefficients seen in Figure 3b mainly reflect sample/probe roughness (Figure 4) and the compressibility of the materials,

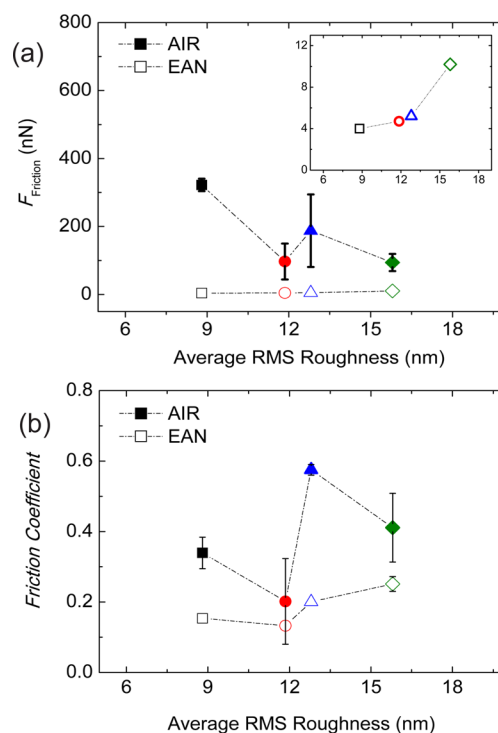


Figure 4. Effect of surface roughness on (a) friction force magnitude at 35 nN applied load and (b) friction coefficient for the four systems studied (from left to right, silica–silica, silica–PTFE, alumina–silica, and alumina–PTFE) in air (closed symbols) and EAN (open symbols). The inset in panel a shows the same EAN data plotted on an enlarged y axis.

as described in a previous publication.⁵⁷ The friction coefficients obtained in EAN were nonetheless lower than the values in air (Figure 3b); the friction generated at the contact of the surfaces in air was reduced by replacing the contact with IL molecules, which nullified the adhesion. As such, the friction reflects rather the properties of the intervening medium, which is capable of constant replenishment.

Figure 4 shows representative friction values (chosen at an arbitrary applied normal load of 35 nN) and friction coefficients versus rms roughness (values shown in Table 1). In order to understand the effect of roughness on friction, a “combined” rms roughness was estimated by arbitrarily averaging the roughness of the probe and that of the substrate. In air, the relationship is nonmonotonic since in addition to changes in roughness there are significant differences in the chemistry and polarity of the contact. PTFE is known to be a low friction material and also precludes the possibility of capillary condensation due to its lack of polarity. This tends to reduce

both the friction coefficient and friction force, accounting for the “kinks” in the air data in Figure 4 (top curves, closed circles and diamonds). The issue of capillary condensation is further complicated by the fact that it is also dependent on the surface roughness^{37,54,57} such that complete flooding of the contact zone can only occur if the roughness is smaller than the Kelvin radius.^{54,57,65} The contention of the last paragraph (that in EAN the frictional dissipation is characteristic of the IL properties, rather than dependent on the chemistry of the respective materials) is strongly borne out by the observation that both friction magnitude and friction coefficients increase monotonically with roughness in EAN.

A number of authors have previously examined the nanotribology of various ILs, generally looking at nm-thick IL films adsorbed on model surfaces. Bhushan et al.^{28,39} evaluated the performance of imidazolium cation-based ILs on silicon surfaces. Their nanoscale friction coefficients obtained from AFM with a Si_3N_4 tip were in the range of 0.02–0.05. Recently, Wang et al.⁷² used AFM with a silicon cantilever to perform measurements on another group of imidazolium ILs finding slightly higher values of $\mu \approx 0.04$ –0.1. Mo et al.^{43,44} used colloid probe AFM with a silica probe to evaluate the performance of imidazolium type ILs on silicon surfaces (and modified silicon surfaces) and again determined similar values of $\mu \approx 0.09$ –0.13. In all of these studies, the lubricity of imidazolium ILs was higher than the values in EAN found here, $\mu \approx 0.10$ –0.25. This might indicate that the imidazolium is a better lubricant, though caution should be applied here in that friction is a system parameter and the roughnesses of the surfaces in this study are higher (how roughness affects friction is still a matter of study,⁷³ as is the role of the probe size on the friction coefficient obtained).⁷⁴ As a comparison, Nainaparampil et al.⁴ have seen larger friction coefficients for pyridinium and pyrrolidinium cation-based ILs studied by colloid probe AFM. Friction coefficients measured for the completely immersed system of a borosilicate probe and a silicon wafer were $\mu \approx 0.3$ –0.8. Given the immaturity of the field, the range of substrates studied, the differences in ILs used, and the fact that roughness data is seldom adequately provided, it is difficult to compare the results presented here with other studies. The focus on surface forces, as opposed to lubricating properties, appears to be unique to this work.

When ball-on-disk measurements were used to measure the lubrication properties of water⁷⁵ or perfluoropolyether lubricants³⁹ (such as Z-TETRAOL), friction coefficients were generally in the range of 0.8 and 0.2–0.6, respectively. These systems all exhibited attractive, repulsive, and adhesive normal forces to varying degrees (as a result of van der Waals, electrostatic, hydrophobic/hydrophilic, bridging forces, etc.) which, as shown here, are not present when metals, metal oxides, and polymer materials are immersed in ILs.

Evidently, it is difficult to separate the contributions of IL chemistry, surface roughness, and experimental approach in order to find the most effective lubricating IL. However, all of these recent studies concur that ILs are considerably better lubricants than water and are promising when compared to other lubricants,^{39,76} particularly when their high thermal stability and low vapor pressure are considered.

As mentioned above, previous work focuses on adsorbed thin films of ILs,^{43,44,72,77} whereas in our experiments, the entire system (probe + substrate) was immersed in the IL. A potential problem with thin films is that they may locally be depleted, exhibiting a gradual increase in μ and they may also be more

subject to water adsorption from the atmosphere.⁷⁸ If the films become depleted due to repeated shearing, physical contact of the materials becomes likely and tribocharging may occur.^{39,78}

When the probe and substrate are immersed in EAN which is a highly viscous liquid ($\eta = 32 \text{ mPa}\cdot\text{s}$),⁵ hydrodynamics plays a key role and the enhanced viscosity may well affect elastohydrodynamic lubrication. Experiments were thus performed in order to clarify the role of hydrodynamics on the normal forces in EAN, by varying the approach rate of the probe toward the surface. On approach, normal force measurements showed a relatively long-ranged repulsive force that increased with increasing approach speed (Figure 5). This

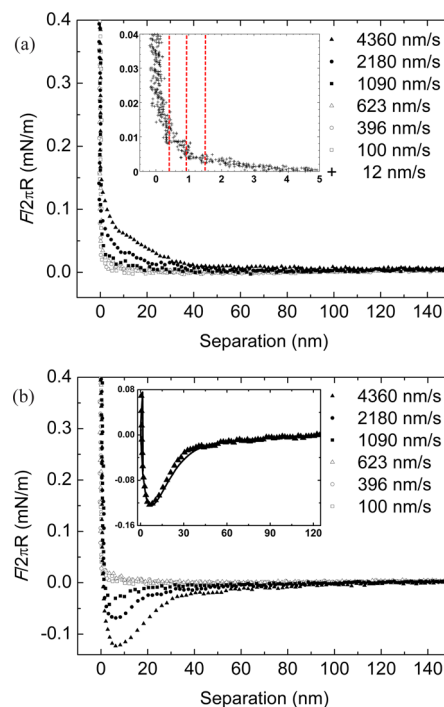


Figure 5. Effect of scan speed on the normalized forces on (a) approach and (b) retraction between a silica probe and a silica surface in EAN. The inset in panel a shows steps characteristic of the physical dimensions of the EAN ion pair at close separations. Such oscillations are only evident at extremely slow scan speeds of $\leq 12 \text{ nm/s}$. The inset in panel b represents the fit to the data obtained upon separation at 4360 nm/s .

apparent repulsion arises due to hydrodynamic resistance in squeezing the liquid EAN out of the contact zone and is more pronounced when the surfaces are moved toward each other quickly. This interaction was first experimentally quantified by Chan and Horn,⁷⁹ who showed that theoretical predictions were followed down to molecular separations. Equation 5 describes the hydrodynamic force as a function of the probe radius, R , separation, h , relative velocity, v , and viscosity, η

$$F = \frac{6\pi R^2 v \eta}{h} \quad (5)$$

At high velocities the hydrodynamic drag is larger than any other minor short-range forces that may be present and this repulsive force provides an additional barrier to the achievement of intimate molecular contact of the surfaces, thus precluding stiction. Conversely, an attractive “mirror image” interaction occurs on separation due to viscous resistance to flow into the contact zone.

We note that hydrodynamics are minimized by taking force measurements at slow rates over small separation distances; under such conditions it is possible to see steps in the normal force profile which correspond to structural layering of the EAN ion pairs (Figure 5a, inset) as shown previously.^{8–10,12,31} In the force vs separation curve obtained at 12 nm/s between a silica probe and a silica surface in EAN, characteristic structural steps were seen, separated by 0.5 nm (red dashed lines), corresponding to the size of the ethylammonium-nitrate ion pair.⁸ This kind of force profile indicates that EAN can still exhibit a structural transition from sponge-like to lamellar layering under confinement despite the local surface roughness, though the resolution and extent of this effect is less pronounced than in ideal, model systems.⁸ There is still considerable debate as to whether the steps are associated with only the flat substrate when an AFM tip is used or whether order layers associated with the tip are also seen. The fact that such layers are also seen with a colloid probe tends to suggest that in fact the layers may be a result of confinement between two surfaces. One argument might be that there is an asperity on the surface acting as an “AFM tip” which causes a similar observation to that in ordinary AFM studies. It is impossible to say exactly how large an area the ordering occurs over, and this is a general phenomenon with AFM force measurement. It must however be more “global” than a single asperity as the results are reproducible in different experiments at slow speeds, and therefore not the result of a single asperity contact in an isolated experiment. This structuring is also an indication that the hygroscopic nature of ILs is not causing significant modification of the interactions. The IL was exposed to the laboratory environment; however, it is known that it is not possible to detect steps in the force curves if the water content rises above 5%. The fact that steps were observed confirms that the water content is less than 5%.¹⁰ Under conditions relevant to MEMS and NEMS for example, the applied loads and rates of movement would be such that hydrodynamics would dominate the force interaction in EAN and mask any influence from liquid structuring and probably also the minor roughness dependence seen above.

The drainage experiments in Figure 5 have all been fitted using the established theory.^{79,80} The analysis was also performed assuming that there might be a slip length, i.e., that the shear velocity is not zero at the walls, but the fit could not be improved by invoking a slip length. Thus a further conclusion is that there is no evidence for wall slip in these (layered) systems. What is remarkable is that the fitted viscosity returned from the fits is systematically much lower than the bulk viscosity of EAN at this temperature. The inset in Figure 5b shows an example of a curve fitted with the hydrodynamic theory of eq 5 where the local speed was determined from the sampling rate and the values of the separation. Due to the intrinsic noise of AFM data, the rate of approach calculated pointwise from the raw data varies dramatically from point to point, rendering fitting impossible. Since the approach rate in reality is monotonic this issue can be easily solved by smoothing the separation (for speed calculations only) by a running average and then fitting it with a polynomial. It can be seen that the model describes the measured data extremely well, however the viscosity returned by the fit is 12 mPa·s, which is an almost 3-fold reduction compared to the bulk viscosity. This striking result is interpreted as directly resulting from the lamellar ordering of the IL at the surfaces which is expected to have an intrinsically lower resistance to sliding, due

to the presence of well-defined shear planes. Although the phenomenon is undoubtedly related to shear thinning, the extracted viscosities vary only very weakly with the drainage rate, so we ascribe the result primarily to “surface thinning”. (Note that very rough surfaces can induce an underestimation of the hydrodynamic minimum due to a significant film thickness at the point of closest approach. However, since the entire curve can be fitted, and the fitted viscosity is also independent of the rate of approach/separation such effects are not present here.) The fact that a single value of the viscosity can be used in the fit is unexpected – one would envisage that the viscosity would vary as a function of separation. However, in this geometrical configuration, at any separation (the value of the closest point on the sphere to the surface) the force is generated by drainage throughout the gap, i.e., at a wide range of separations. Thus it may well be that the average value extracted in fact disguises an even lower viscosity for the lamellar region exemplified by the inset to Figure 5b. Such a surface thinning behavior is likely to have positive implications in reducing viscous losses at high shear rates in lubricating ILs where the intrinsic boundary lubricating properties provide benefit at low loads. Such an effect was predicted, but not substantiated, in earlier work, where it was noted that silica particle suspensions in EAN settled much more rapidly than predicted by the hindered Stokes equation.¹⁰

■ CONCLUSIONS

For the first time, the relation between the lateral and normal forces measured by colloid probe AFM has been demonstrated with different materials of industrial relevance immersed in an IL. The major effect of EAN is to completely screen all the conservative surface forces, including the adhesive forces ordinarily seen in liquids, van der Waals forces, electrostatic double-layer forces, and capillary forces (which is a major component of the adhesion in air and which may play a role in water and some oil-based lubricants as well). Due to this screening, the forces that lead to stiction are effectively eliminated and the presence of IL at the interface leads to enhanced lubricity. The complex dependence of the friction on material properties is largely removed when the normal surface forces are screened by EAN since the contact largely reflects the properties of the EAN, mediated slightly by the surface roughness. The friction behavior becomes much more uniform, with only a weak dependence on the surface roughness appearing to distinguish the different surface combinations. This is despite the fact that the choice of materials included both positive and negative Hamaker constants based upon refractive indices. Such an abolition of normal forces is a result of the molten salt nature of the IL and is expected to be a general result which should be independent of the IL composition as well as the chemistry of the interacting surfaces.

EAN is a viscous liquid and hydrodynamics thus become significant as the surfaces change their relative separation. There is no evidence for wall slip in EAN. From a tribological perspective this resistance to movement within EAN adds a further barrier to the achievement of contact, precluding adhesion and ensuring low friction. The large hydrodynamic resistance however is likely to impede the use of ILs, at least in immersion, for NEMS and MEMS where such forces are expected to be determining. While at higher approach speeds the force curves were featureless, steps due to structural layering in EAN were observed with a characteristic step size of 0.5 nm at slow approach speeds indicating that the beneficial

and spontaneous layering of the IL under confinement is measurable, even for relatively rough surfaces.

Finally, this layering has led to dramatically reduced viscosity under confinement which is a finding so far unique to this work. This result is potentially far reaching and could impact upon the use of ILs in any application involving confinement, such as in nanofluidic devices and dye solar cells, and the potential to tune lubricating properties via a combination of boundary lubrication (direct adsorption) and “surface thinning” opens up new possibilities for lubricant design in general.

■ ASSOCIATED CONTENT

● Supporting Information

SEM picture of an AFM colloidal probe and other force curves and friction measurements discussed in this work. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

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